

FUNCHAL, Portugal

WMO: 085220

Lat: 32.6475N

Lon: 16.8925W

Elev: 56

StdP: 100.65

Time Zone: 0.00 (EUW)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	12.7	13.3	1.5	4.3	19.3	3.6	4.9	16.8	7.1	18.3	6.2	18.3	1.7	20	0.213

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.2	27.7	20.7	26.8	20.7	26.1	20.5	22.3	26.0	21.8	25.5	21.3	25.0	2.3	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.0	15.8	24.6	20.4	15.2	24.5	19.9	14.7	24.3	65.5	26.2	63.7	25.6	62.1	25.0	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.5	4.6	3.9	11.4	31.3	1.0	2.4	10.6	33.0	10.0	34.4	9.5	35.7	8.7	37.5		
(5)			7.4	23.3	0.7	0.5	6.9	23.7	6.4	23.9	6.0	24.2	5.5	24.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	19.9	17.2	16.8	17.3	17.8	19.0	20.9	22.6	23.7	23.4	22.1	19.9	18.2		
	DBStd	2.77	1.35	1.39	1.49	1.38	1.39	1.13	1.10	1.14	1.06	1.40	1.61	1.27		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	178	40	47	38	23	7	0	0	0	0	0	4	18		
	CDD10.0	3621	222	190	227	236	279	327	389	424	401	376	296	254		
	CDD18.3	757	4	3	7	9	29	77	131	166	151	118	50	13		
	CDH23.3	1761	1	2	15	20	23	55	288	607	477	231	36	6		
	CDH26.7	118	0	0	0	5	6	2	16	50	25	13	1	0		
(14) Wind	WSAvg	1.6	1.8	1.8	1.9	1.7	1.5	1.3	1.2	1.2	1.4	1.6	1.8	1.9		
(15) Precipitation	PrecAvg	617	90	77	63	39	20	9	2	2	23	88	98	97		
	PrecMax	1508	300	479	251	209	108	68	32	17	192	247	306	296		
	PrecMin	368	2	0	0	0	0	0	0	0	0	0	3	2		
	PrecStd	210	79	79	53	40	22	16	5	4	37	66	68	80		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.2	23.2	25.3	26.4	25.7	25.9	28.5	30.7	28.7	28.2	26.3	24.1		
		MCWB	15.4	13.8	13.4	15.3	17.0	20.5	20.8	18.9	21.0	18.1	16.3	13.9		
	2%	DB	21.1	21.2	22.3	22.3	23.3	24.7	26.5	27.6	27.2	26.3	24.2	22.1		
		MCWB	15.8	15.0	15.0	16.3	18.2	19.7	20.8	21.7	21.1	20.2	18.4	16.5		
	5%	DB	20.3	20.2	20.9	21.0	22.4	23.8	25.6	26.7	26.5	25.4	23.2	21.1		
		MCWB	15.4	14.9	15.6	16.0	17.7	19.2	20.4	21.3	20.8	19.9	18.1	16.4		
	10%	DB	19.5	19.3	19.9	20.2	21.5	23.1	24.8	26.0	25.7	24.6	22.4	20.4		
		MCWB	15.0	14.4	15.2	15.5	17.1	18.7	20.0	20.9	20.4	19.5	17.7	16.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.2	17.7	17.9	18.2	19.7	21.2	22.2	23.1	23.1	22.5	21.1	19.0		
		MCDB	20.4	19.7	20.6	21.6	23.0	24.9	26.3	27.0	26.5	25.5	23.4	20.9		
	2%	WB	17.3	16.7	17.1	17.4	18.8	20.4	21.5	22.4	22.2	21.6	20.2	18.2		
		MCDB	19.8	19.4	19.8	20.8	22.3	24.0	25.5	26.4	25.7	24.6	22.8	20.5		
	5%	WB	16.6	15.9	16.5	16.8	18.2	19.7	21.0	21.8	21.6	21.0	19.2	17.5		
		MCDB	19.4	19.0	19.5	20.2	21.5	23.2	24.7	25.9	25.3	24.2	22.2	20.2		
	10%	WB	15.8	15.2	15.9	16.2	17.6	19.2	20.5	21.4	21.1	20.3	18.4	16.7		
		MCDB	18.9	18.6	19.1	19.6	20.9	22.6	24.2	25.3	24.8	23.7	21.6	19.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.7	5.0	5.2	5.0	4.9	4.6	4.8	5.2	5.2	5.1	4.8	4.5		
		MCDBR	5.4	5.9	6.3	6.0	5.8	5.4	5.7	6.0	6.0	5.8	5.4	5.2		
	5% WB	MCWBR	4.2	4.5	4.4	4.2	4.0	3.6	3.7	3.7	3.9	4.1	4.0	4.1		
		MCWBR	4.9	5.2	5.4	5.5	5.4	5.0	5.2	5.6	5.5	5.1	4.9	4.8		
(40) Clear-Sky Solar Irradiance	taub	taub	0.341	0.362	0.389	0.378	0.381	0.380	0.389	0.393	0.390	0.375	0.366	0.347		
		taud	2.552	2.473	2.388	2.436	2.451	2.477	2.455	2.466	2.491	2.526	2.511	2.546		
	Ebn at Noon	Ebn at Noon	880	892	889	911	906	902	892	886	877	867	842	851		
		Edh at Noon	83	99	115	114	114	110	112	110	103	92	86	80		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	2.86	3.68	4.93	6.05	6.64	7.09	7.08	6.61	5.55	4.22	3.04	2.56		
	RadStd	RadStd	0.14	0.25	0.25	0.22	0.41	0.34	0.29	0.22	0.18	0.18	0.18	0.13		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page